METRIC

MILC-768A(OS)
28 February 1990
SUPERSEDING
JAN-C-768(OS)
4 May 49

MILITARY SPECIFICATION

COPPER, POWDERED (FOR USE IN AMMUNITION) (METRIC)

This specification is approved for use by the Naval Sea Systems Command Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes one grade of powdered copper.

1.2 Part number. Part number for cataloging purposes under this specification is coded as follows:

MO0768	X	XX
Specification	Granulation	Purity

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N\A FSC 1370

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STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111 -5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-11 Standard Specification for Wire-Cloth Sieves for Testing Purposes (DoD Adopted)

(Applicatition for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Attn: Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification, Rules and Regulations

(Application for copies should he addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General material requirements. Powdered copper shall satisfy the requirements given herein and shall be made from electrolytic copper.

3.2 Total copper. Minimum, 99.0 percent by weight.

3.3 Granulation. Through No. 200 Standard sieve, minimum, 99.5 percent by weight.

3.4 Workmanship. The material shall be uniform in quality and free from foreign materials that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Lot size. The maximum lot size shall be 227 kilograms (kg).

4.3 Sampling. Three containers shall be selected, at random, from the lot for sampling purposes. If there are fewer containers in the lot, all containers shall be sampled. A 250-gram(g) sample representative of each container shall be placed in a clean, dry, rubber-stoppered bottle. The samples shall be thoroughly mixed to form a composite sample representative of the lot. The bottle shall be labeled to show the name of the material, manufacturer, plant, number of the contract or order, number of kilograms in the lot, and the lot number.

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4.4 Tests,

4.4.1 Total copper. Transfer a weighed portion of approximately 2 g of the sample to a 200-milliliter(ml) electrolytic beaker. The weight of the specimen shall be determined on an analytical balance to the nearest 0.001g. Add 20 ml of 50-percent nitric acid solution and boil until the copper is dissolved and all oxides of nitrogen have been driven off. Add 10 ml of 50-percent sulfuric acid, dilute the solution to about 150 ml, and electrolyze. Use sand-blasted platinum electrodes, the cathode being tared, and a current density of about 3 amperes per 100 square centimeters of cathode surface. When the deposition of the copper from the solution is complete, remove the electrolyte and immediately wash the cathode with distilled water. Rinse the cathode twice with ethyl alcohol, then burn off the excess alcohol while moving the electrode back and forth through the air. Cool the cathode in a desiccator and weigh. Calculate the increase in weight as percent total copper.

4.4.2 Granulation. Attach a bottom pan to a U.S. Standard No. 200 sieve conforming to the requirements of ASTM E-11. Place 100g of the sample on the sieve, cover, and shake for 5 minutes by hand or mechanically by means of a shaker geared to produce 800 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh any material remaining on the sieve and calculate the percentage passing through the sieve.

4.43 Inspection of packaging. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shaIl be in accordance with MIL-STD- 129.

4.4.4 Lot acceptance or rejection. When the results of the tests in 4 show that the requirements of 3 and 5 have been met, the lot represented by the sample shall be accepted. Otherwise, the lot shall be rejected.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.2 Packing. Unless otherwise specified (see 6.2), the material shall be packed in a container having a capacity of 4.54 kg and so constructed as to ensure acceptance by common or other carrier, for safe transportation, at the lowest rate to the point of delivery. Containers used shall comply with National Motor Freight Classification Rules and the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Marking shall be in accordance with MIL-STD 129. Marking shall include, but not be limited to, the following:

a. Number and revision letter of this specification
b. Manufacturers name

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C. Material designation
d. Lot number
c. Date of manufacture
s. Net weight
g. Purchase order number

6. NOTES

(This **section** contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6,1 Intended use. This material is intended for use in pyrotechnic compositions.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation **and,** if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Preservation level, if required (see 5.1).

d. Special packing or marking (see 5.2 and 53).

63 **Subject term (key word) listing.**

Electroanalysis
Granulation
Purity

6.4 **Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - OS
Project No.(1370-N288)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL C-768A(OS)	2. DOCUMENT DATE (YYMMDD) 2 April 1990

3. DOCUMENT TITLE
COPPER, POWDERED (FOR USE IN AMMUNITION) (METRIC)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commanding Officer Naval Ordnance Station	b. TELEPHONE (Include Area Code) (1) Commercial: (301)743-4510 (2) AUTOVON: 364-4510
c. ADDRESS (Include Zip Code) Code 3730 Indian Head, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198-230

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